25900387 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCFSQ201 - FOOD CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Gelatinization of starch granules results. [U] / [CO1]

   a). The process where starch granules dissolve in water

   b). The disruption of molecular order within starch granules

   c). The process where starch granules crystallize

   d). The binding of amylose to water molecules

2. Identify the disaccharide also called "milk sugar". [U] / [CO1]

   a). Maltose

   b). Lactose

   c). Isomaltose

   d). Trehalose

3. Enzyme resposible for browning of fruits and vegetables. [U] / [CO1]

   a). Amylase

   b). Lipase

   c). Polyphenolase

   d). oxidase

4. In a Fischer projection of D- glucose the OH group on the penultimate carbon (C5) is [K] / [CO1]

a). on the left b). on the right

c). above the plane d). below the plane

5. An example of a chemical agent causing protein denaturation [U] / [CO2]

a). heat b). freezing

c). Isopropyl alcohol d). None of these

6. Solution with low pH implies that it is [U] / [CO2]

a). Basic b). Neutal

c). Acidic d). None of these

7. Fibrous protein are soluble in [U] / [CO2]

a). Water b). Acetone

c). Both of them d). None of these

8. Recognize the type of lipid in lecithin [K] / [CO3]

a). Glycolipid b). Phospholipid

c). Sterol d). Waxes

9. Identify the effect of hydrogenation on iodine value. [U] / [CO3]

a). Decreases iodine value b). No effect

c). Depends on acid value d). Increases iodine value

10. Identify the type of rancidity caused by enzymatic hydrolysis of fats. [U] / [CO3]

a). Autoxidation b). Oxidative rancidity

c). Reversion d). Hydrolytic rancidity

[1x10 = 10]

Part B

Short Answer

Answer any **4** questions

Each question carries **5** marks

11. Write a note on the classification of monosaccharides [K] / [CO1]

12. Define the terms a) water holding capacity b) water binding capacity c) water activity [K] / [CO1]

13. Write the structures of amino acid with R groups as aliphatic hydrocarbon. [U] / [CO2]

14. Discuss the determination of Michaelis constant (Km) and Vmax. [K] / [CO2]

15. Explain how RM value helps in detecting adulteration in butter. [U] / [CO3]

16. Describe the difference between natural and synthetic antioxidants. [U] / [CO3]

[5x4 = 20]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

17. Explain the structure, composition, properties and industrial uses of starch and pectin. [U] / [CO1]

18. Elaborate the structural classification of carbohydrate with examples [K] / [CO1]

19. Explain in detail on protein structure [U] / [CO2]

20. Describe the classification of fatty acids. [U] / [CO3]

[10x2 = 20]